Midterm

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(30 points)* Given matrix $A = \begin{bmatrix} 1 & 0 & a \\ 0 & -1 & b \\ 1 & 0 & c \end{bmatrix}$.

a) Determine the values of a, b, and c that make matrix A a full-rank matrix, and provide a justification for your chosen values.

b) Is matrix A invertible? If it is, find the inverse; if it isn't, provide an explanation.

c) Find a basis for $C(A)$ (the column space of A)?

d) What is the dimension of $N(A)$ (the nullspace of A)?

Question 2 *(25 points)* Given matrix $A = \begin{bmatrix} 3 & 1 & 5 & 9 & -1 \\ 1 & 0 & 1 & 2 & -1 \\ -2 & 1 & 0 & -1 & 2 \\ 1 & 1 & 3 & 5 & 0 \end{bmatrix}$ and vector $b = \begin{bmatrix} 11 \\ 5 \\ -8 \\ 4 \end{bmatrix}$,

a) Solve the system of linear equations $Ax = b$.

b) Find the rank of A.

c) Find the nullspace of A.

Question 3 *(10 points)* Can any vector in R^3 be written as a linear combination of columns of $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & 1 \\ -1 & 2 & 1 \end{bmatrix}$.

Question 4 *(20 points)* Given matrix $A = \begin{bmatrix} 0 & 2 & 1 \\ 16 & 4 & -6 \\ -16 & 4 & 10 \end{bmatrix}$. Check whether the vectors $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ are eigenvectors of the matrix A. If yes, find their corresponding eigenvalues.

Question 5 *(15 points)* Find the angle between the vector $x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and the y-axis.